

UST KAMENOGORSK, Kazakhstan

WMO: 350675

Lat: 50.037N

Lon: 82.494E

Elev: 286

StdP: 97.93

Time Zone: 6.00 (E06)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-39.4	-36.2	-43.0	0.1	-39.4	-39.8	0.1	-36.5	9.5	-2.2	8.7	-3.2	0.5	10	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	33.0	18.7	31.1	18.2	29.2	17.8	20.7	28.4	19.8	27.6	19.2	26.9	3.7	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.5	23.4	17.2	12.7	22.4	16.2	11.9	21.6	60.7	28.3	57.8	27.6	55.4	26.8	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.3	7.3	DB	-40.5	35.5	3.7	1.8	-43.1	36.8	-45.3	37.8	-47.4	38.8	-50.0	40.1	(4)
(5)			WB	-40.5	23.1	3.6	1.4	-43.2	24.1	-45.3	25.0	-47.3	25.8	-50.0	26.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.4	-17.7	-16.7	-5.4	7.5	12.9	19.1	20.2	18.5	12.4	5.6	-4.4	-13.0	(6)	
	DBStd	15.02	9.69	9.06	7.56	5.52	4.54	3.68	2.77	3.21	4.55	4.83	8.04	9.11	(7)	
	HDD10.0	3537	860	749	478	107	23	1	0	0	26	149	432	713	(8)	
	HDD18.3	5654	1118	982	737	326	175	36	13	37	183	395	681	971	(9)	
	CDD10.0	1112	0	0	0	31	114	274	317	264	98	13	1	0	(10)	
	CDD18.3	187	0	0	0	0	8	59	72	43	6	0	0	0	(11)	
	CDH23.3	3087	0	0	0	31	223	867	951	781	224	12	0	0	(12)	
	CDH26.7	1139	0	0	0	3	57	331	364	307	77	1	0	0	(13)	
(14) Wind	WSAvg	2.6	2.0	1.9	2.4	3.0	3.6	2.8	2.6	2.5	2.4	2.7	3.0	2.5	(14)	
(15) Precipitation	PrecAvg	472	26	25	28	31	50	43	52	45	35	43	43	41	(15)	
	PrecMax	828	84	82	84	89	106	124	214	162	99	114	83	89	(16)	
	PrecMin	322	8	0	5	3	0	12	5	4	1	12	3	6	(17)	
	PrecStd	111	15	17	20	18	29	30	38	31	21	23	19	24	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.8	3.8	13.9	26.9	31.1	34.2	35.1	35.1	32.2	25.2	15.8	5.2	(19)	
		MCWB	0.1	0.9	6.9	12.6	16.7	18.5	19.5	18.3	16.0	12.3	9.4	1.1	(20)	
	2%	DB	1.0	1.9	8.9	23.9	28.0	32.1	32.2	32.8	28.8	21.1	10.9	3.0	(21)	
		MCWB	-0.9	-0.4	5.0	12.2	15.0	18.5	19.4	18.4	14.7	11.1	5.8	0.7	(22)	
	5%	DB	-1.0	-0.2	6.1	21.1	25.8	30.2	30.2	30.1	25.8	17.2	7.9	1.2	(23)	
		MCWB	-2.6	-1.7	3.2	11.0	14.2	18.1	19.2	17.9	14.2	9.4	4.6	-0.8	(24)	
	10%	DB	-3.1	-3.0	4.1	18.1	23.2	28.2	28.8	28.0	22.2	14.1	5.0	-0.9	(25)	
		MCWB	-4.4	-4.2	1.9	10.1	13.3	17.9	18.7	17.3	12.9	8.4	2.9	-2.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.0	1.4	8.0	14.5	18.7	21.4	22.6	21.5	17.6	13.9	10.0	2.0	(27)	
		MCDB	2.5	2.9	12.2	23.2	28.9	28.7	29.9	29.3	28.3	21.6	14.1	3.6	(28)	
	2%	WB	-1.0	-0.3	5.1	13.1	16.6	20.4	21.0	19.9	15.8	11.6	6.5	0.7	(29)	
		MCDB	1.1	1.5	7.9	21.0	23.8	28.0	28.6	28.2	25.4	20.0	9.6	2.7	(30)	
	5%	WB	-2.9	-2.1	3.5	11.9	15.3	19.6	20.0	19.0	14.7	10.1	4.5	-0.6	(31)	
		MCDB	-1.3	-0.5	5.8	19.3	23.0	27.3	27.4	27.3	23.4	16.7	7.2	1.2	(32)	
	10%	WB	-4.6	-4.3	2.1	10.5	14.2	18.7	19.3	18.1	13.6	8.7	3.0	-2.5	(33)	
		MCDB	-3.3	-3.2	3.9	17.4	21.6	26.4	26.7	25.8	21.2	13.2	4.8	-0.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.2	14.0	12.0	14.5	15.9	15.4	14.9	16.7	16.4	12.6	9.3	10.6	(35)	
		MCDBR	11.6	12.9	12.4	20.9	20.0	19.1	18.9	21.3	22.2	20.1	12.1	9.6	(36)	
	5% WB	MCWBR	10.1	11.2	8.8	10.6	9.7	8.0	8.3	8.8	10.1	11.3	8.0	7.7	(37)	
		MCWBR	10.5	12.6	11.3	18.0	15.9	15.8	17.5	19.1	17.9	11.0	9.3	7.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.304	0.342	0.395	0.433	0.420	0.419	0.440	0.407	0.367	0.368	0.343	0.311	(40)	
	taud		2.022	1.949	1.933	2.057	2.155	2.235	2.185	2.257	2.314	2.202	2.076	1.983	(41)	
	Ebn at Noon		686	755	784	806	838	841	814	825	821	732	631	601	(42)	
	Edh at Noon		89	125	153	151	143	133	138	123	104	97	85	80	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.51	2.63	3.96	4.68	5.92	6.40	5.91	5.49	4.04	2.20	1.26	1.09	(44)	
	RadStd		0.10	0.11	0.35	0.44	0.46	0.53	0.47	0.37	0.37	0.28	0.14	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page